ML10 Cleaning without hiding evidence

Week 4, Clean, chart, question | Session 10 of 24 | Machine Learning | New library: pandas

Today's question. What is wrong with this table, and what did I change?

Objective. By the end: you can count missing values and duplicates, remove an exact duplicate on a copy, and save a cleaning log.

> **WHERE THIS IS USED**
> Clinics, schools and banks collect data typed by busy people, so duplicates and blanks are normal. Data scientists clean the table and write down every change they make. What would you check first in a messy register?

Before this session: ML09 Mean, median, range and proportions. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
missing value	A cell with no recorded value. Its true value is unknown.	A blank in the price column.
duplicate	A row repeated exactly. It may be an accident or a real second sale.	The same receipt copied twice.
raw data	The file as you received it. Keep it unchanged.	The original receipt book.
cleaning log	A list of every change made and why.	The marking notes beside a corrected exercise.
to_csv	Saves a table into a CSV file.	Copying the register to a new page.
index=False	Tells to_csv not to write the extra row-number column.	Not copying the page numbers.

Block A. Count the problems

A tiny table has two sets of trouble. Count them before you change anything.

Predict. Predict how many missing prices and how many duplicate rows the raw table has. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
raw = pd.DataFrame({"id": [1, 2, 2, 3], "price": [1000, None, None, 3000]})
missing = raw["price"].isna().sum()
dupes = raw.duplicated().sum()
clean = raw.drop_duplicates().copy()
```

```
print(missing, dupes, len(clean))
print(clean["price"].mean())
```

Run it, then compare. The cell prints 2, 1 and 3, then 2000.0. isna marks each blank as True and sum counts them. duplicated marks the repeated row and drop_duplicates removes it. copy makes a separate table, so raw stays unchanged. The mean of 2000.0 uses the two prices that exist and ignores the missing one.

Investigate.

1. Why do we work on a copy?
2. Why not replace the missing price with 0?

Modify. Print how many prices are still missing in clean.

Block B. Save a cleaning log

Write down what you found as a small table and save it as a file.

Predict. Predict how many rows and columns the saved log has after you read it back. Write your guess on paper first.

Your guess, written before you run it

```
log = pd.DataFrame({"step": ["raw rows", "duplicate removed", "price still missing"],
    "count": [4, 1, 1]})
log.to_csv("cleaning_log.csv", index=False)
log_back = pd.read_csv("cleaning_log.csv")
print(log_back)
print(log_back.shape)
```

Run it, then compare. The log has 3 rows and 2 columns. to_csv writes the table into cleaning_log.csv. index=False stops pandas adding a column of row numbers. Reading the file back proves that it saved. The log lists the raw row count and each change, so a reader can repeat the cleaning.

Investigate.

1. What would the log look like with index=False left out?
2. Who is the cleaning log for?

Modify. Add the step final rows with a count of 3 and save again.

Make

Write a cleaning log for this table: four raw rows, one removed duplicate and one remaining missing price. Save it as my_log.csv and read it back. Explain why replacing the missing price with 0 would change its meaning.

Save your work as ML10. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
FileNotFoundError when reading the log back	to_csv did not run, or ran in another folder.	Run the to_csv line again and check the file list in Jupyter.
KeyError: 'price'	The column name has a different spelling in this table.	Print list(raw.columns) and copy the exact name.

Exit ticket

Cleaning keeps the raw data safe, counts each problem, and writes down every change in a log.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.